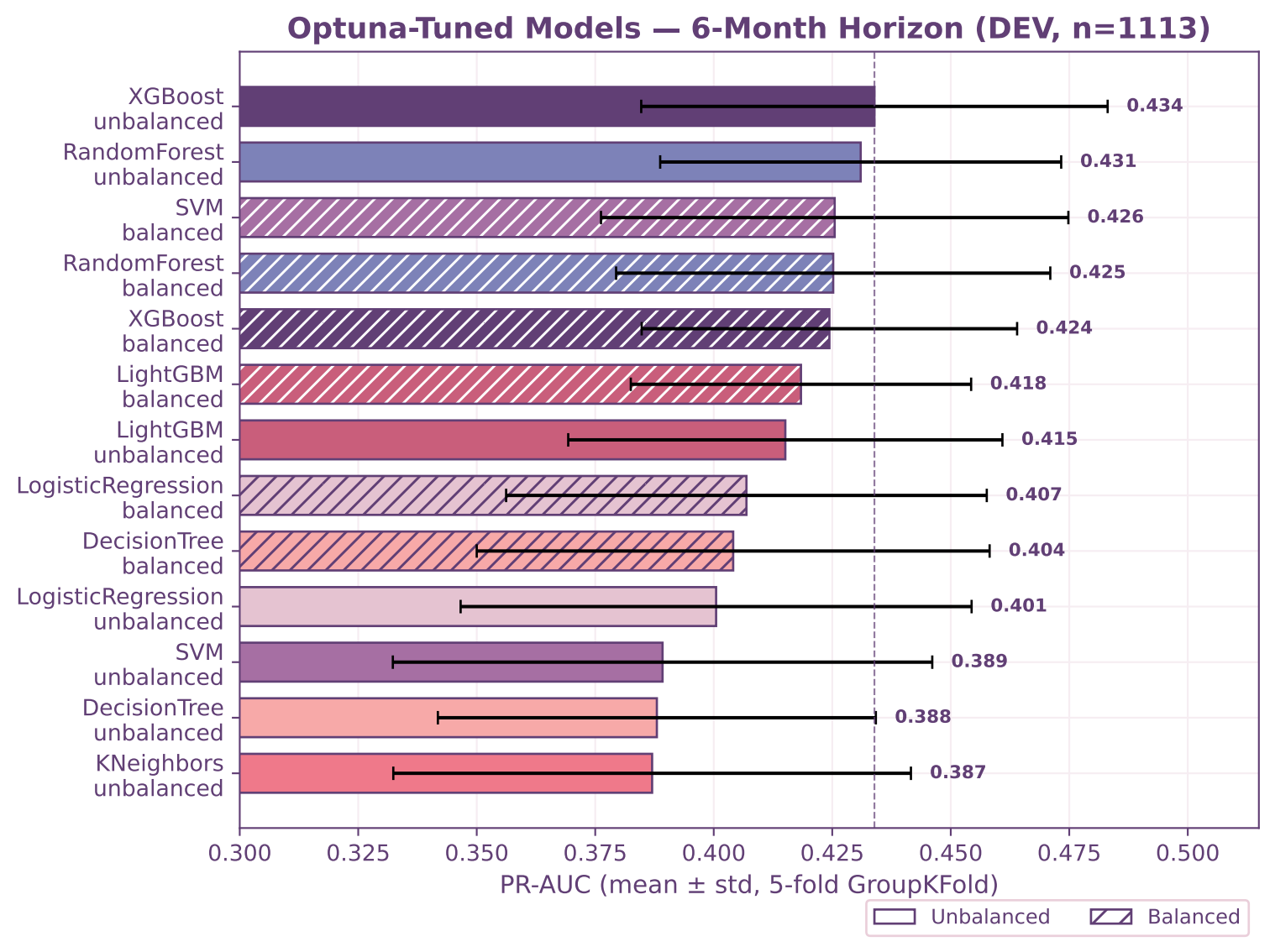
Optuna-Tuned Models — 6-Month Horizon (DEV, n=1113)
XGBoost unbalanced
RandomForest unbalanced
SVM balanced
RandomForest balanced
XGBoost balanced
LightGBM balanced
LightGBM unbalanced
LogisticRegression balanced
DecisionTree balanced
LogisticRegression unbalanced
SVM unbalanced
DecisionTree unbalanced
KNeighbors unbalanced
0.434
0.431
0.426
0.425
0.424
0.418
0.415
0.407
0.404
0.401
0.389
0.388
0.387
0.300
0.325
0.350
0.375
0.400
0.425
0.450
0.475
0.500
PR-AUC (mean ± std, 5-fold GroupKFold)
Unbalanced
Balanced